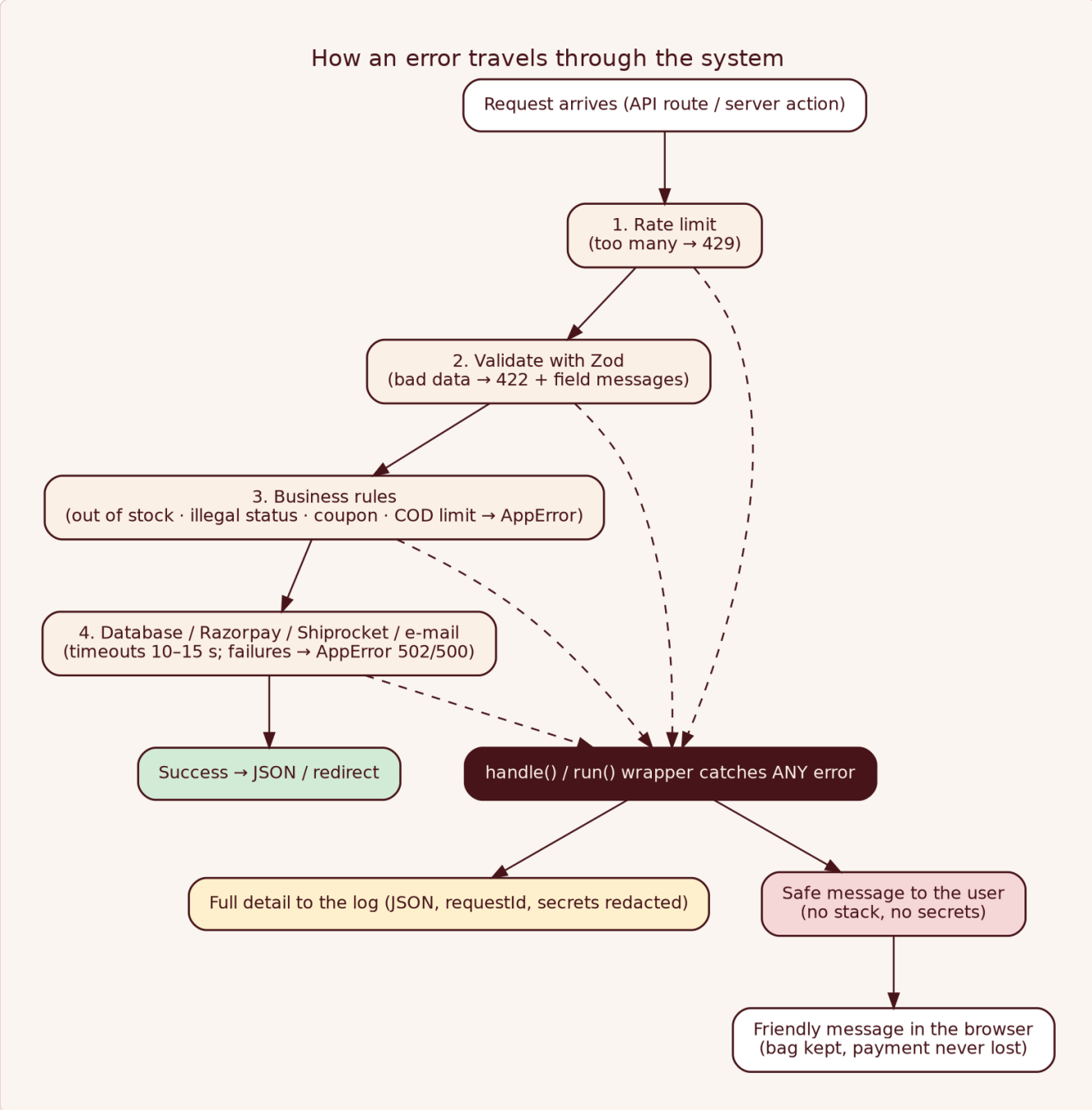
LUXE & THREAD · DOCUMENTATION

INTERNAL – for developers

Internal 16 · Error Handling

Audience	INTERNAL – developers



1 · Principles

1. **Validate at the door** (Zod → 422 with per-field messages). 2. **One error type** `AppError(code, message, status)`. 3. **One catch-all wrapper**: API routes use `handle()`, admin actions use `run()`; both log full detail and return a safe message. 4. **Customers see friendly text; logs hold the truth.** 5. **Never lose a payment.** 6. **Everything repeatable is idempotent.**

2 · Standard error response

```
{ "error": { "code": "VALIDATION_ERROR", "message": "Only 2 left of Kasavu Set (M).",
"details": { "code": "INSUFFICIENT_STOCK", "available": 2 }, "requestId": "5b1c…" } }
```

Quote `requestId` → `grep` it in the logs.

3 · Codes

Code	HTTP	Meaning	Typical cause / action
`VALIDATION_ERROR`	400/409/422	bad input or business rule	read message (stock, coupon, COD limit, PIN)
`NOT_FOUND`	404	order/product/link token wrong	
`UNAUTHORIZED`	401	wrong webhook/job token	check secrets
`RATE_LIMITED`	429	too many requests	wait
`INVALID_STATE`	409	action not allowed in this status	e.g. ship an unpaid order
`INVALID_SIGNATURE`	400	payment/webhook signature mismatch	key/secret mix-up
`CONFIG_MISSING`	503	env value missing	fill env
`PAYMENT_PROVIDER_ERROR`	502	Razorpay down/rejected	see logs `providerMessage`
`SHIPPING_PROVIDER_ERROR`	502	Shiprocket down/rejected	API user, wallet, pickup name
`STORAGE_ERROR`	500	database error (`sqlState` in details)	Troubleshooting §4
`INTERNAL_ERROR`	500	unexpected	stack trace in log

4 · Admin actions

`run(path, okMessage, fn)` redirects back with `?ok=` (green) or `?err=` (red). Known `AppError`s show their message; unknown errors show "See the server log". Redirect "errors" (`NEXT_REDIRECT`) are re-thrown so navigation works.

5 · Scenario table

Scenario	Behaviour
Last piece bought elsewhere	order transaction rolls back; "just sold out"

Scenario	Behaviour
Razorpay down at checkout	nothing saved or reserved; "please try again"
Browser closes after paying	webhook marks PAID; order link works
verify fails after paying	no error shown; order page auto-refreshes until PAID
Duplicate webhooks	idempotent
Payment after cancellation	auto-refund
Shiprocket down	order stays PAID; error saved; retry button
Courier cancel fails	order unchanged; `CANCEL_BLOCKED`; customer told
SMTP down	outbox retries 5× with back-off; then `failed` + admin retry
Corrupt browser bag	sanitised; server re-prices
DB down	shop shows "could not load"; `/api/health` 503
Page crash	`error.tsx` friendly page; `global-error.tsx` last resort
Disk/permission error on photo upload	red message; photo not attached

5b · Logging

One JSON line per event with `level`, `message`, `requestId`, `code`, `status`; secrets redacted. Read with `pm2 logs`. Levels: info/warn/error.

6 · Adding error handling to new code

Validate with Zod · throw `AppError` for expected failures · wrap routes in `handle()` · timeouts on every outside call · make it idempotent · add a unit test.